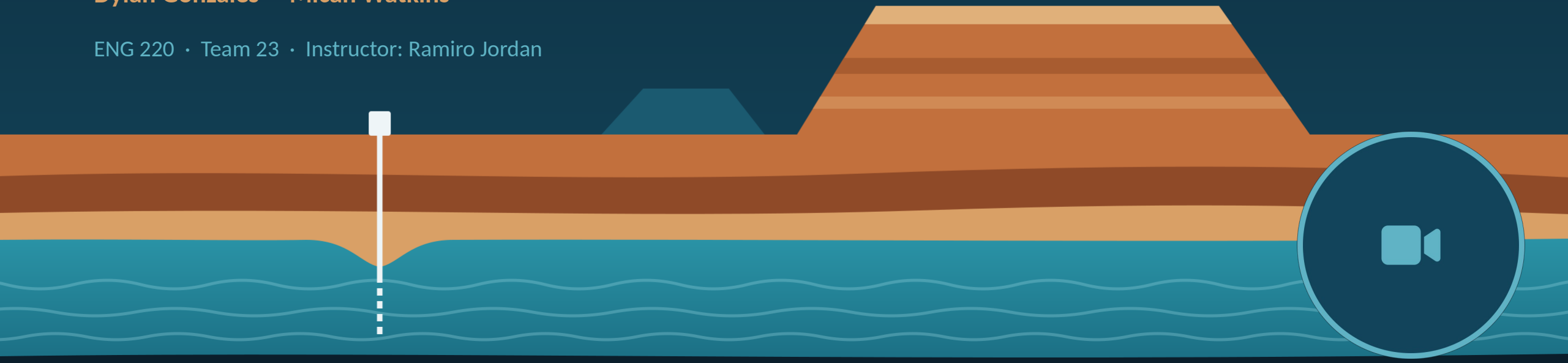
Beneath the Mesa

Assessing Aquifer, Well, and Acequia Water Resources Across Sandoval County, New Mexico

Petar Djordjevic (PM) · **Meckrem Yousef** · **James Wanueka**
Dylan Gonzales · **Micah Watkins**

ENG 220 · Team 23 · Instructor: Ramiro Jordan



Background

- Where Sandoval County's water comes from: wells, rivers, and acequias
- Why the county spans several aquifer systems
- The concern: declining groundwater near Cuba
- What this project set out to check



Figure caption and source

Research questions

MAIN QUESTION

Write the project's main question in one sentence

1. **Groundwater levels**
   Write the question in one sentence, using the wording from the proposal.

2. **Cuba's aquifers**
   Write the question in one sentence, using the wording from the proposal.

3. **Water quality and acequias**
   Write the question in one sentence, using the wording from the proposal.

Data

Groundwater levels

Source, years covered, and number of sites

Aquifer tests

Source, years covered, and number of sites

Water quality

Source, years covered, and number of sites

Acequias

Source, years covered, and number of sites

Full source links are on the last slide and in the project documentation.



Methods

Collect

Dylan Gonzales

One or two lines on what this step did

Clean

James Wanueka

One or two lines on what this step did

Analyze

Micah Watkins

One or two lines on what this step did

Present

Meckrem Yousef

One or two lines on what this step did

Report

Petar Djordjevic

One or two lines on what this step did

Tools Python · pandas · Plotly · Streamlit · GitHub

Data collection

Dylan Gonzales

Data acquisition and dataset justification

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

A data source page or the raw dataset

Figure caption and source

Data cleaning

James Wanueka

Data cleaning and preprocessing

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

The cleaning script, or the data before and after

Figure caption and source

Analysis and visualizations

Micah Watkins

Analysis code and visualizations

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

A chart from the analysis

Figure caption and source

Streamlit app

Meckrem Yousef

GitHub workflow and the Streamlit app

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

A page of the live app

Figure caption and source

Report and documentation

Petar Djordjevic

Project manager: report, slides, and video

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

The documentation site or a page of the report

Figure caption and source

Results

Chart, map, or app screenshot

Copy this slide for each result

Figure caption and source

— ft

Key number and what it measures

What this shows

Two or three sentences on the main pattern in the chart.

Findings

What the data show

- Measured results, with numbers

What they suggest

- Likely explanations, stated as possibilities

Too limited to tell

- Questions the data can't answer yet

Bottom line
One sentence that answers the main question



Limitations

- Areas with few or no monitoring wells
- Short or irregular water-level records
- Data the project couldn't get
- What a stronger study would need

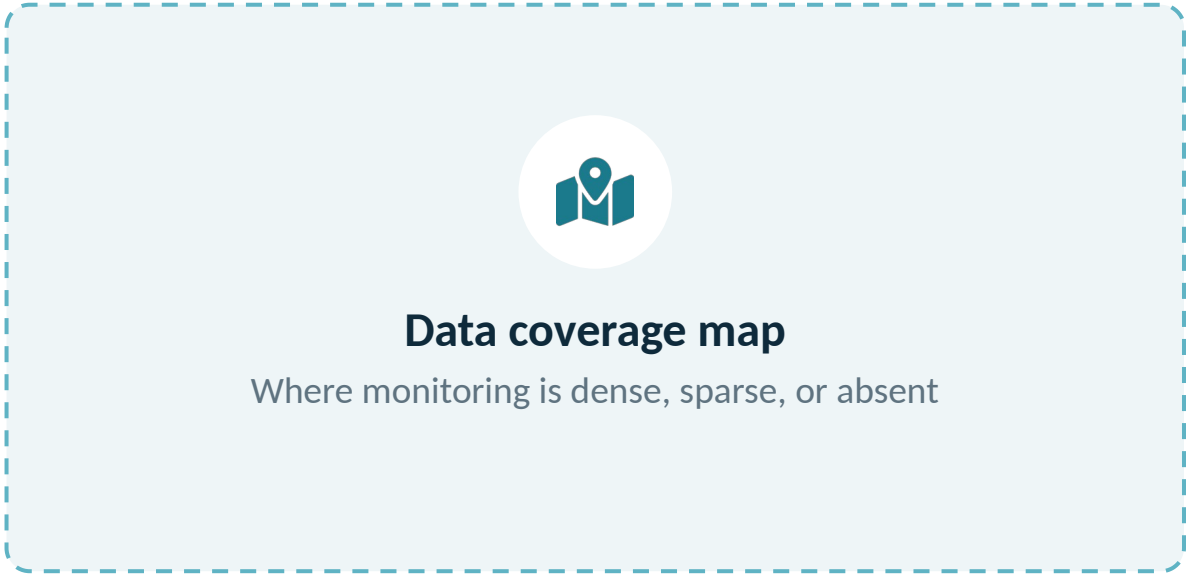


Figure caption and source

Explore the data, code, and app

GitHub repository
github.com/pdjordj11/eng_220_aquifer_project

Streamlit app
eng220aquiferproject.streamlit.app

Documentation
pdjordj11.github.io/eng_220_aquifer_project

Datasets
github.com/pdjordj11/eng_220_aquifer_project/tree/main/data

